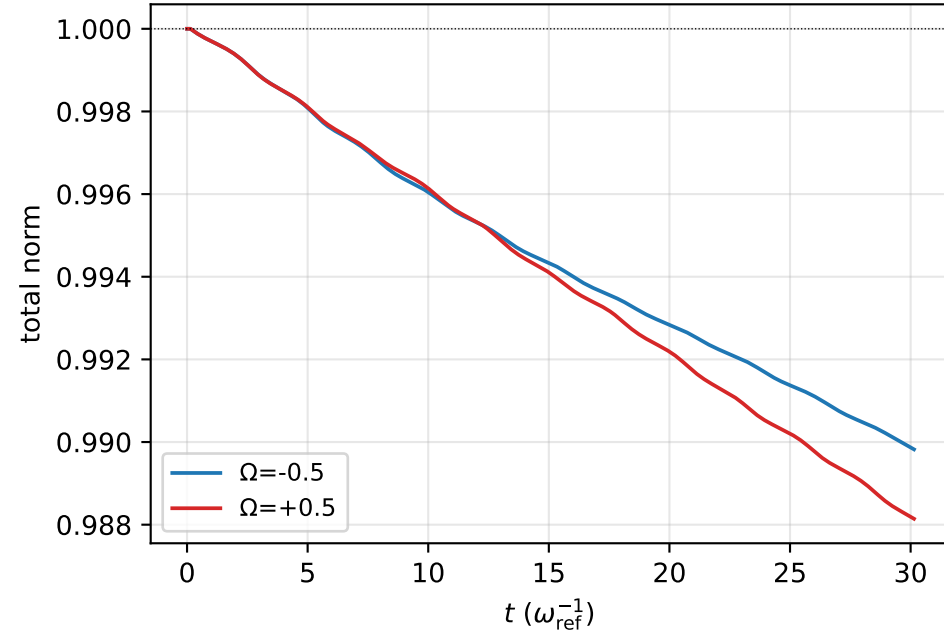
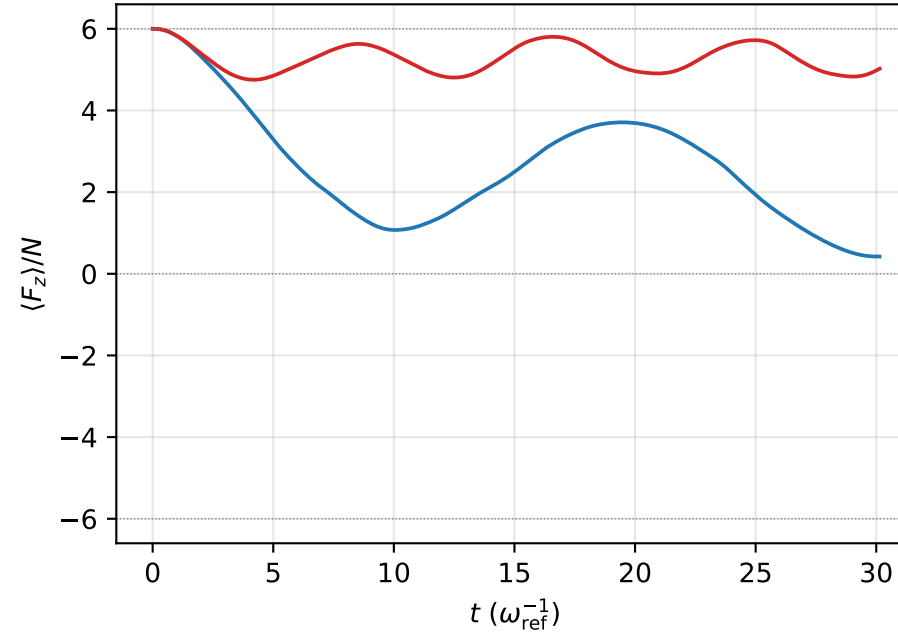


Eu151 Barnett spin pumping — weak Bz (2.6 nT, non-secular) $\times$ Klaus stir $\times$ $\pm\Omega$ + DDI + K3 + γ_{dr}
 32^3 grid, $30 \omega^{-1}$ (~ 43 ms), $\Omega = \pm 0.5$ (adiabatic, $T_{\text{stir}} \approx 12.6 \omega^{-1}$ vs $T_{\text{Larmor}} \approx 9 \omega^{-1}$)

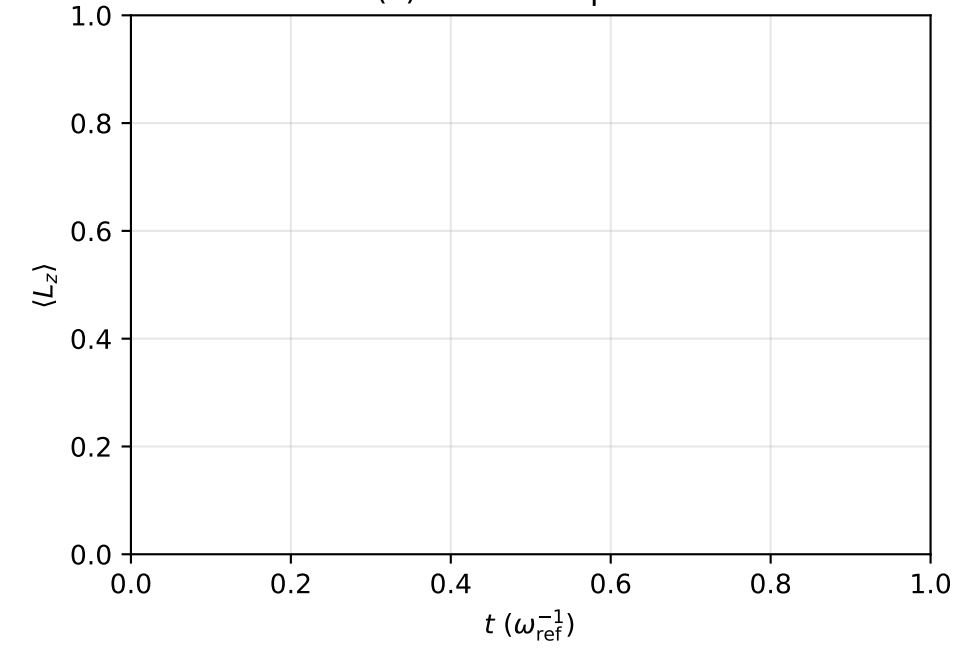
(a) atom-number decay



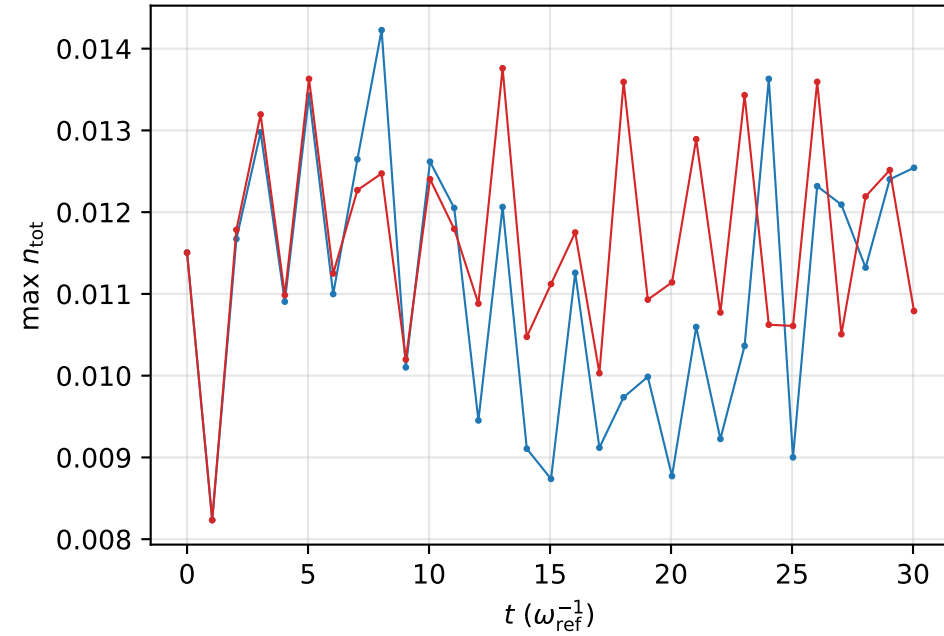
(b) Barnett pumping — headline



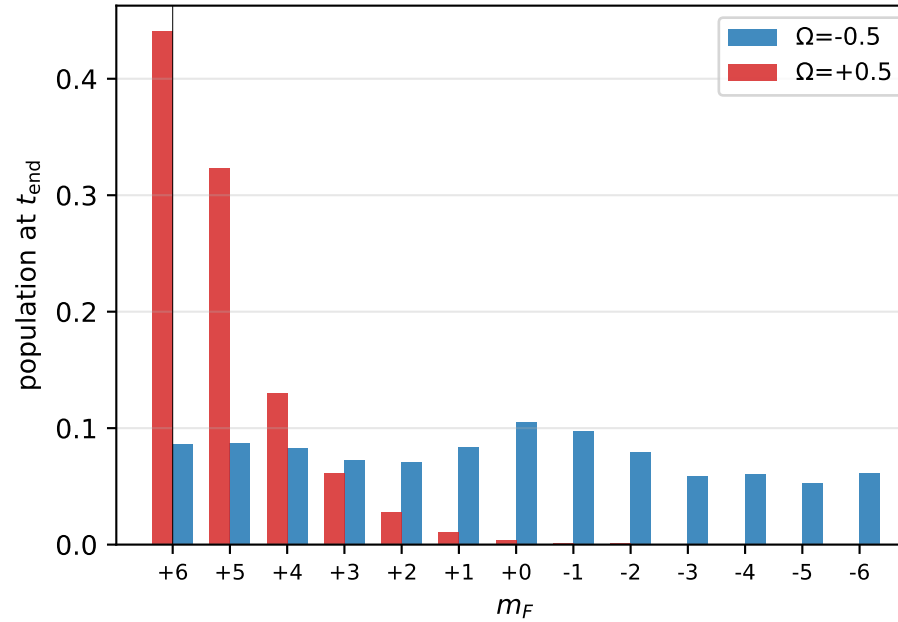
(c) orbital response



(d) peak density



(e) spin distribution at end — Barnett pumping



(f) sign-asymmetry — Barnett antisymmetry

